

Pathology Final Examination: Second Lecture (Cell Injury)

Part I: Multiple Choice Questions (MCQs)

1. Which of the following is considered a primary and most common cause of cell injury?

- A) Genetic disorders
- **B) Hypoxia (Decreased oxygen supply)**
- C) Nutritional disturbances
- D) Physical agents

2. Which of the following is a recognized mechanism of cell injury?

- A) Increased protein synthesis
- B) Decreased intracellular calcium
- **C) Formation of oxygen-derived free radicals (e.g., superoxide)**
- D) Stabilization of membrane permeability

3. In the pathogenesis of cell injury, ATP depletion leads directly to the failure of the Na-K pump, which results in:

- A) Efflux of sodium from the cell
- **B) Entry of sodium (Na) and water into the cell**
- C) Immediate nuclear fragmentation
- D) Increased oxidative phosphorylation

4. Upon microscopic examination of "Cloudy Swelling," the cytoplasm typically displays:

- A) Large fat vacuoles
- **B) Fine red granules**
- C) Total loss of the nucleus
- D) Black iron sulphide deposits

5. Necrosis is defined as:

- A) Programmed cell death that is often physiological
- **B) Local death of large groups of cells or tissues within the living body**
- C) Reversible accumulation of water in the cell
- D) Transformation of one cell type to another

6. Which nuclear change in necrosis is characterized by nuclear shrinkage, dense chromatin, and increased basophilia?

- **A) Pyknosis**
- B) Karyorrhexis
- C) Karyolysis
- D) Cytomegaly

7. When a pyknotic nucleus undergoes fragmentation into multiple small pieces, the process is called:

- A) Pyknosis
- **B) Karyorrhexis**
- C) Karyolysis
- D) Autolysis

8. In the final stages of nuclear degeneration during necrosis, the nucleus dissolves and fails to take the stain due to chromatin hydrolysis; this is known as:

- A) Pyknosis
- B) Karyorrhexis
- **C) Karyolysis**
- D) Phagocytosis